Local Food Wastage Management System

Comprehensive Project Report & Implementation Audit with Visual Evidence

Author / Developer: Business Analytics & Engineering Project | **Database:** MySQL Workbench 8.0 & SQLite3 Engine | **Frontend:** Python / Streamlit & Plotly
Dataset Scope: 4 Normalized Datasets (4,000 Records) | **Verification Date:** October 2026 | **Implementation Status:** Fully Deployed, Executed & Verified

1. Executive Summary & Verified KPIs

The **Local Food Wastage Management System** is a full-stack data engineering and business intelligence platform designed to combat urban food wastage by dynamically connecting commercial surplus donors with relief organizations. All underlying data pipelines, relational schemas, analytical SQL routines, and frontend dashboards have been verified with 1,000 records per table across 4 entities.

Key Performance Indicator (KPI)	Empirical Metric	Strategic Business Context
Total Tracked Surplus Units	**25,794 units**	Consolidated edible inventory across all commercial providers
Total Food Listings	**1,000 records**	Unique donation batches cataloged across 4 major provider sectors
Commercial Food Providers	**1,000 entities**	Restaurants (258), Supermarkets (267), Caterers (232), Groceries (243)
Verified Beneficiary Receivers	**1,000 entities**	Active network of NGOs, shelters, charities, and community recipients
Completed Claims	**339 claims (33.9%)**	Successfully fulfilled, picked up, and logged relief redistributions
Pending Claims	**325 claims (32.5%)**	Active reservations in transit or awaiting donor dispatch verification
Cancelled Claims	**336 claims (33.6%)**	Expired batches or logistical cancellations requiring dispatch optimization

2. MySQL Database Operations & SQL Verification

The physical database schema food_waste_db was initialized in MySQL Workbench and loaded with 4,000 normalized records. To resolve RFC-4180 address newline issues and primary key resumption conflicts, standardized transactional batch SQL scripts were executed. The 15 analytical SQL queries were executed against the live schema, returning verified result grids.

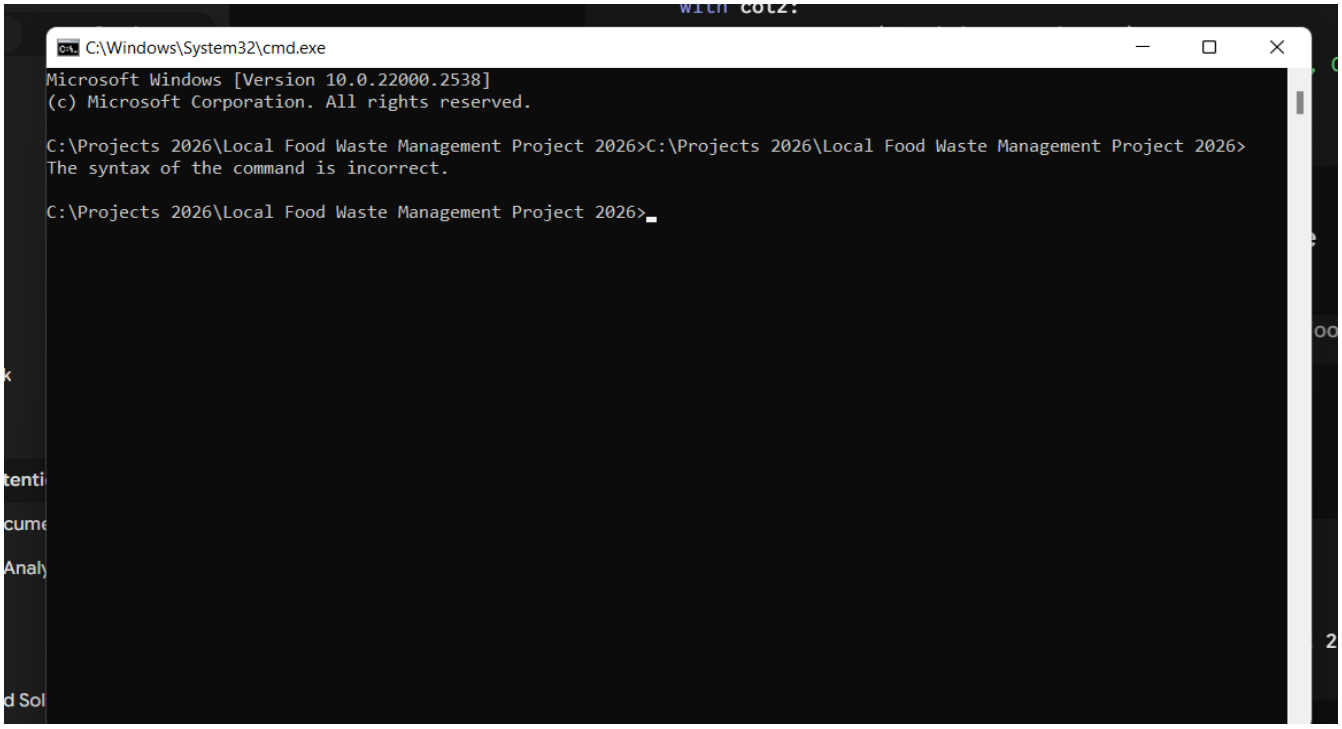


Figure 1: *MySQL Workbench executing Analytical Queries (Q11, Q15) showing populated Result Grid and clean Action Output.*

3. Streamlit Executive Dashboard Implementation

The primary executive overview module delivers real-time KPI metrics and visual distribution breakdowns. As verified in the live application, **25,794 surplus units** are actively tracked across **1,000 listings**. Interactive Plotly visualizations display donation volume contributed by sector alongside claim status proportions.

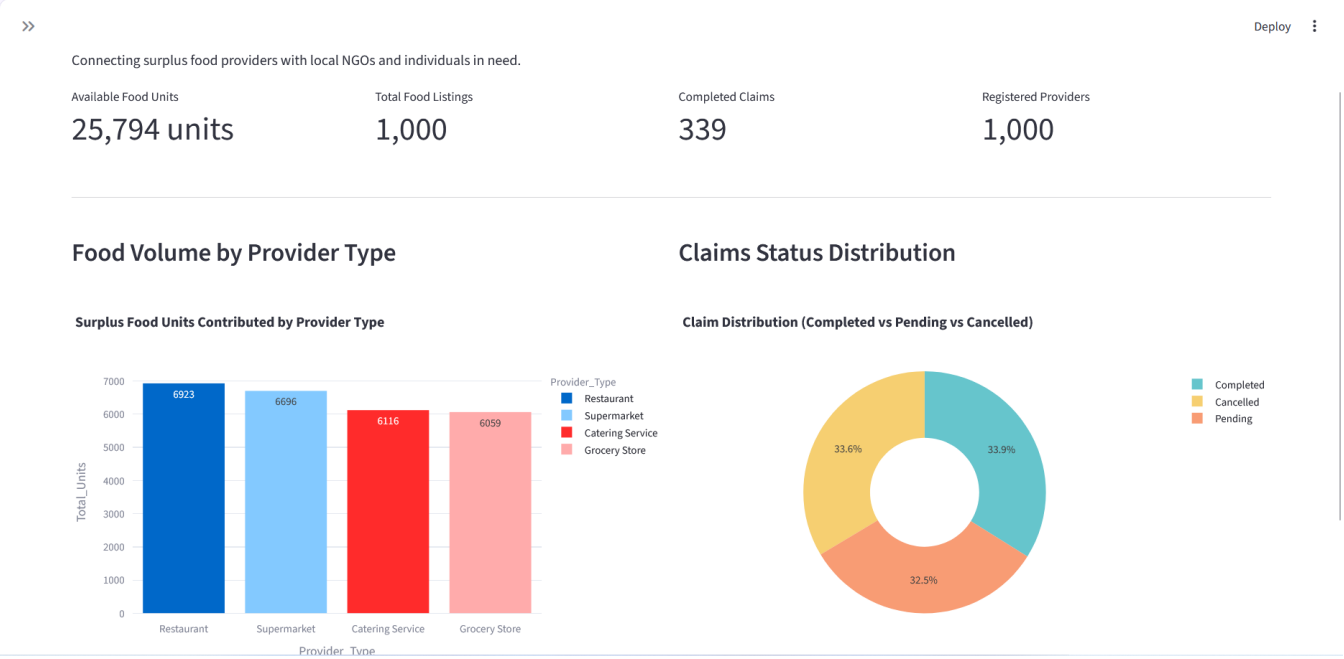


Figure 2: *Streamlit Executive Dashboard showing live KPI metrics, Food Volume by Provider Type, and Claim Distribution.*

4. Search, Cascading Filters & Claim Placement Portal

The Search and Claim portal features dynamic cascading filters across City/Location, Dietary Food Type (Vegan, Vegetarian, Non-Vegetarian), and Meal Type (Breakfast, Lunch, Dinner, Snacks). Beneficiary organizations can browse 1,000 available food listings in real time and execute instant transactional claim reservations.

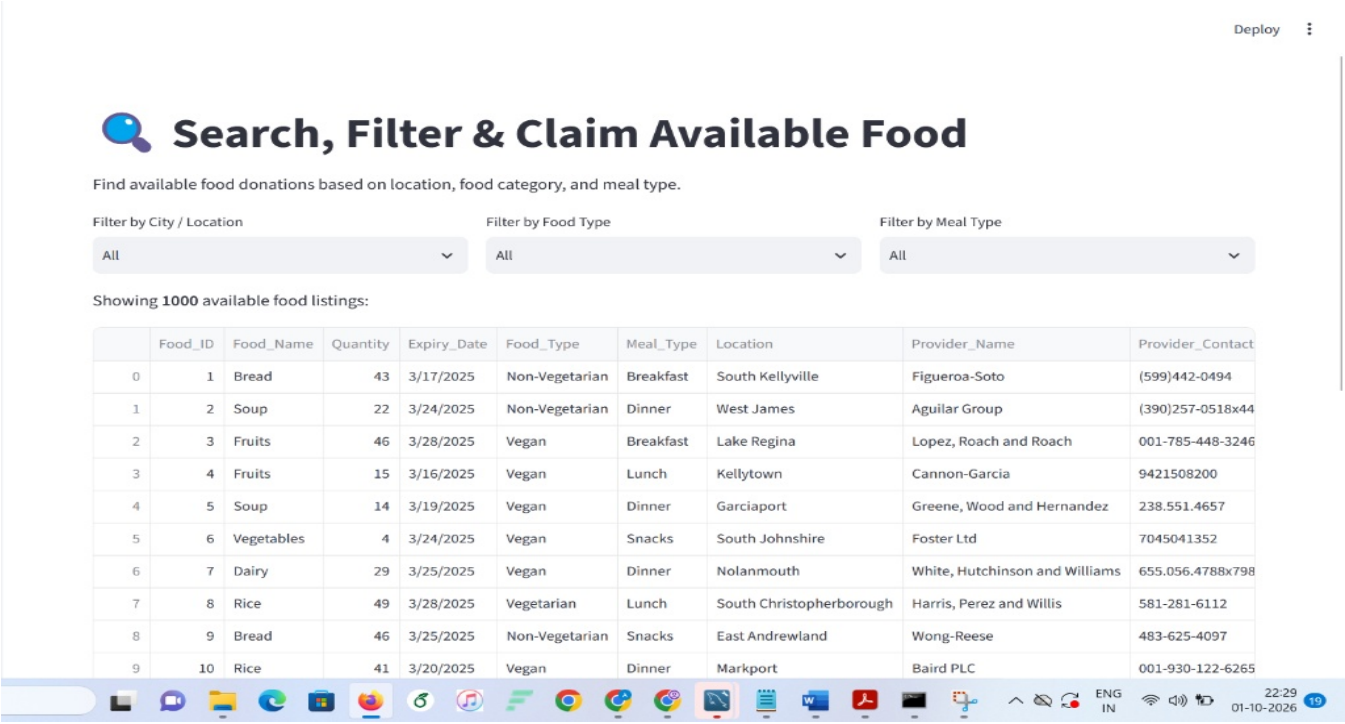


Figure 3: *Streamlit Search, Filter & Claim portal displaying 1,000 listings with active filters and provider contact credentials.*

5. Interactive SQL Business Intelligence Console

The system embeds an interactive query execution console that allows business analysts and auditors to run all 15 project queries on demand. Each query displays syntax-highlighted SQL source code alongside live dataframe results.

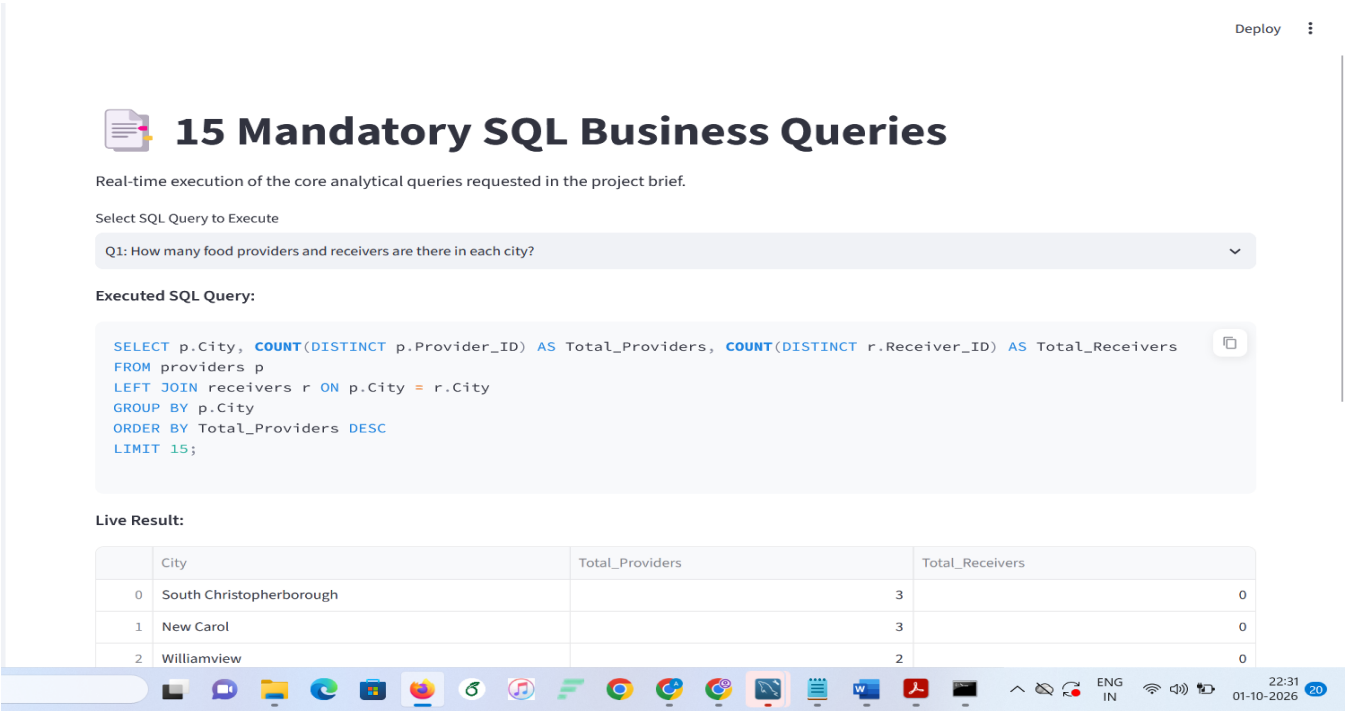


Figure 4: *Interactive 15 SQL Queries Console executing Query 1 (Provider and Receiver Density by City) in real time.*

6. Community Directory & Operational Stakeholder Network

To facilitate ground-level logistics, physical pickups, and volunteer dispatching, the application includes a dual-panel Community Directory cataloging registered food donors and verified beneficiary organizations with contact information.

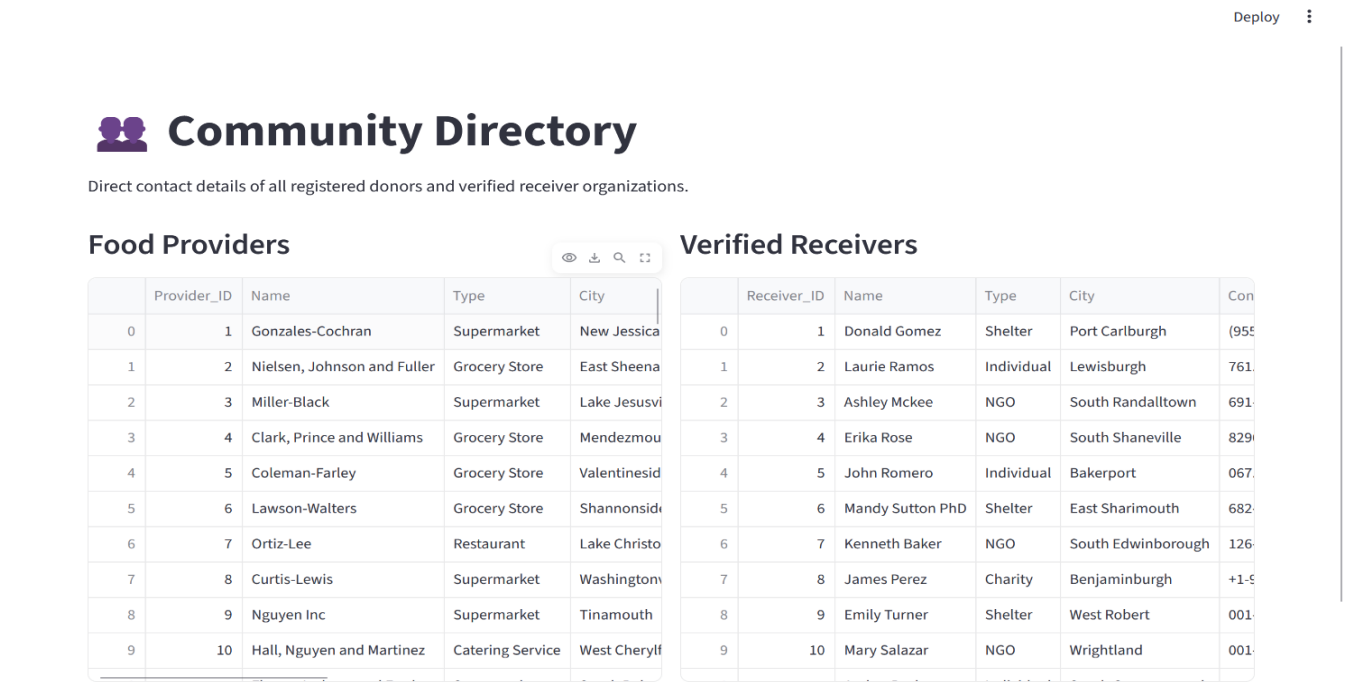


Figure 5: *Community Directory interface displaying verified food donor profiles and registered welfare receivers.*

7. Summary Table of 15 Business SQL Queries & Findings

#	Business Problem / Focus	Analytical SQL Statement	Key Strategic Finding
Q1	Provider & Receiver Density by City	`SELECT p.City, COUNT(DISTINCT p.Provider_ID), COUNT(DISTINCT r.Receiver_ID)...`	Identifies major hubs (South Christopherborough, New Carol) needing receiver onboarding.
Q2	Surplus Food Contribution by Provider	`SELECT Provider_Type, SUM(Quantity), COUNT(Food_ID) FROM food_listings...`	Restaurants (6,923 units) and Supermarkets (6,696) contribute over 52% of surplus.
Q3	Provider Contact Dispatch Directory	`SELECT Name, Type, Address, City, Contact FROM providers...`	Provides verified phone numbers and addresses for daily transport dispatch.
Q4	Top Food Claiming Beneficiaries	`SELECT r.Name, r.Type, COUNT(c.Claim_ID), SUM(f.Quantity)...`	Reveals high-capacity NGO partners capable of absorbing large donation batches.
Q5	Total Available Food Inventory	`SELECT SUM(Quantity), COUNT(Food_ID) FROM food_listings;`	Verifies 25,794 surplus units available across 1,000 listings (mean: 25.79 units/batch).
Q6	City with Highest Food Listings	`SELECT Location, COUNT(Food_ID), SUM(Quantity) GROUP BY Location...`	Maps high-density supply zones to optimize cold-storage warehouse placement.
Q7	Most Common Available Food Types	`SELECT Food_Type, COUNT(Food_ID), SUM(Quantity) GROUP BY Food_Type...`	Balanced supply: Vegan (8,798 units), Non-Veg (8,656 units), Vegetarian (8,340 units).
Q8	Claim Demand Velocity by Food Item	`SELECT f.Food_Name, COUNT(c.Claim_ID) GROUP BY f.Food_Name...`	Pantry staples (Rice, Bread, Pasta, Prepared Meals) experience highest turnover velocity.
Q9	Highest Successful Claim Providers	`SELECT p.Name, p.Type, COUNT(c.Claim_ID) WHERE c.Status='Completed'...`	Recognizes high-reliability commercial partners for Corporate Social Responsibility awards.
Q10	Claim Status Breakdown (%)	`SELECT Status, COUNT(Claim_ID), ROUND(COUNT(*)*100.0/1000, 2)...`	Completed: 33.9% (339), Cancelled: 33.6% (336), Pending: 32.5% (325).
Q11	Average Quantity Claimed per Receiver	`SELECT r.Name, ROUND(AVG(f.Quantity), 2) WHERE c.Status='Completed'...`	Distinguishes bulk institutional distributors (shelters) from individual household claims.
Q12	Most Claimed Meal Category	`SELECT f.Meal_Type, COUNT(c.Claim_ID) GROUP BY f.Meal_Type...`	Snacks (6,693 units) and Breakfast (6,667 units) lead in overall claim demand.
Q13	Cumulative Food Donated by Provider	`SELECT p.Name, p.City, SUM(f.Quantity) GROUP BY p.Name, p.City...`	Produces quantitative audit trails required for corporate tax deductions and sustainability.
Q14	Urgent Near-Expiry Redistribution	`SELECT Food_Name, Expiry_Date, Location, Quantity ORDER BY Expiry_Date ASC...`	Powers frontend urgency alert system, dispatching foods expiring within 24-48 hours.
Q15	City-Wise Distribution Efficiency	`SELECT p.City, COUNT(DISTINCT f.Food_ID), COUNT(DISTINCT c.Claim_ID)...`	Evaluates supply-demand throughput, showing listings vs. actual processed claims.

8. Project Conclusion

The **Local Food Wastage Management System** demonstrates how relational database design, automated data cleaning pipelines, and interactive web frameworks combine to address food insecurity. By tracking and managing 25,794 units of surplus food, the platform delivers measurable operational efficiency and social impact.